

GET THE RENOVATION PLANNING GUIDE

The villa renovation planning guide

Plan your renovation with confidence.

Avoid the hidden mistakes that affect your home's comfort, water performance and long-term running costs.

Comfort

Strong showers, every floor

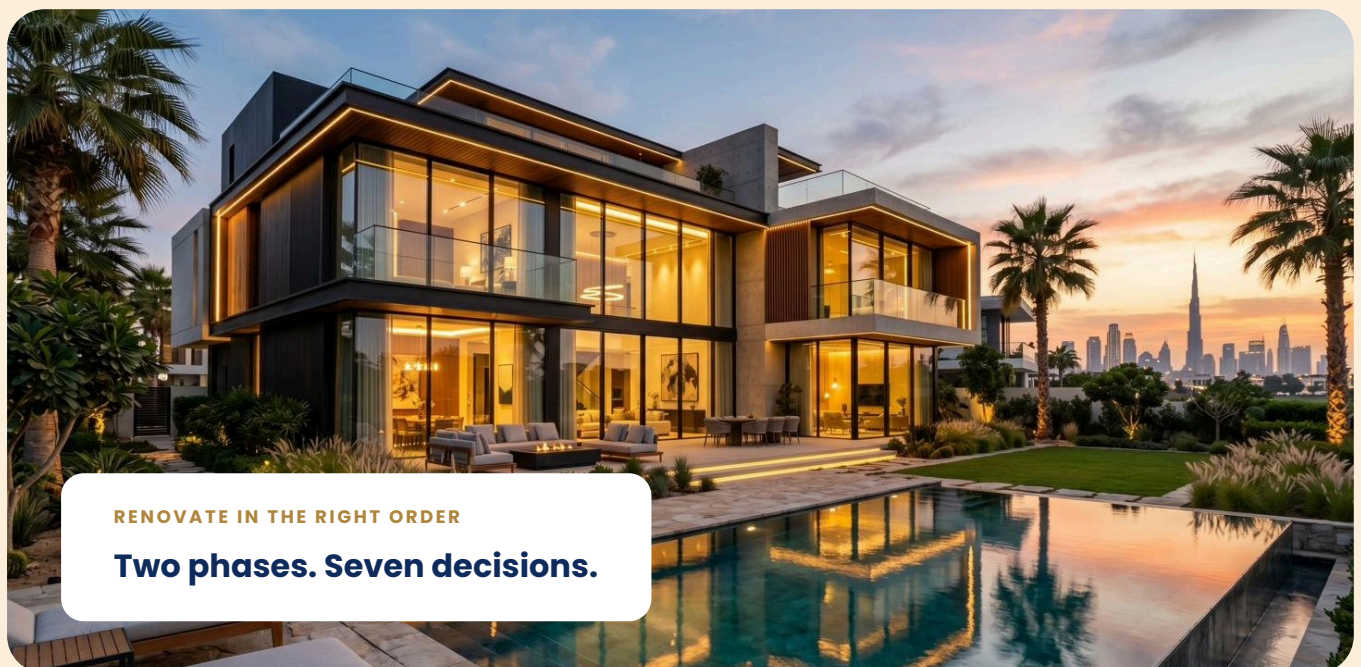
Water performance

Pressure that holds at peak

Running costs

No rework after handover

Inside: a two-phase readiness protocol, a water pressure audit you can run yourself, a decision tree, the mistakes to avoid, and the seven decisions that shape everyday comfort long after the finishes are chosen.



RENOVATE IN THE RIGHT ORDER

Two phases. Seven decisions.

Decisions before demolition

Renovating a villa is a sequence of decisions, and the order matters. Everything below is cheap to decide now and expensive to revisit once the walls are open.

YOUR RENOVATION TIMELINE



DECISION REQUIRED — BEFORE RENOVATION STARTS

Tick each one before the first invoice is signed. Anything unticked is a decision your contractor will make for you.

- | | |
|---|--|
| ☐ Have I finalized bathroom layouts? | ☐ Will multiple bathrooms operate simultaneously? |
| ☐ Do I know my water demand? | ☐ Have I discussed the booster system with my contractor? |

STAGE 5 . STRUCTURAL & UTILITY READINESS, TWO WEEKS OUT

01 DEWA & main utility handover

Confirm every account is fully active in your name. Request a meter read so you have a clean baseline for your first consumption cycle.

02 Central AC & HVAC stabilization

Run the air conditioning at 21 degrees for a continuous 48 hours. Check the vents for condensation leaks and confirm the smart thermostats are calibrated.

03 Deep cleaning & sealing

Schedule deep cleaning before anything is unpacked. Seal natural stone, marble counters and bathroom grouting while the surfaces are still bare.

EXPERT TIP FROM THE ENGINEERS

Never assume new means functional. Run your utility stress tests before the moving truck arrives, so minor snags do not turn into day-one disruptions.



PHASE 1 OF 2

Next, the water pressure audit

The multi-zone stress test

A larger home challenges water in three ways at once: vertical height to the upper floors, long pipe runs across the plot, and several people drawing water at the same time. Run this protocol before you commit to finishes, then record your result below.

STEP 1 . MULTI-FLOOR SIMULTANEOUS TEST

Pressure under vertical load

Go to the highest floor and open the master suite shower. While it runs, have someone open two taps on the ground floor, for example the kitchen sink and a guest bathroom.

AUDIT POINT

Does the upper-floor shower drop, flicker, or lose temperature stability?

STEP 2 . LANDSCAPE & APPLIANCE CONFLICT TEST

Pressure under competing demand

Start the garden irrigation or a high-demand appliance such as the washing machine. Then open the master bathroom shower.

AUDIT POINT

Does your rain shower fade to a weak trickle?

YOUR RESULT — TICK ONE

Record what you saw at the taps. Each outcome means something different for your plan.

☐ **Pressure stable**

WHAT IT MEANS

Flow held steady at every outlet. Your system copes with today's demand.

DO THIS

Re-test once your new fixture list is final — a rain shower changes the maths.

☐ **Minor drop**

WHAT IT MEANS

Noticeable but usable dip. The system is already near its limit.

DO THIS

Add pressure to the contractor brief now. Extra bathrooms will push it over.

☐ **Significant drop**

WHAT IT MEANS

Your infrastructure cannot hold peak demand. Finishes will not fix this.

DO THIS

Specify a variable-speed booster before any bathroom work begins.

CHECKED YES TO ANY DROP?

Your infrastructure cannot hold peak demand.

IS PRESSURE PLANNING CRITICAL FOR YOU?

Water pressure decision tree

Answer these simple questions about your renovation to understand if pressure planning is critical.

EXPERT TIP

Pressure issues are almost always discovered after construction is complete. Plan early to avoid costly changes later.

1 Planning a large rain shower or multiple showers?

YES

High water demand

Rain showers use two to three times more water than standard showers.

NO ↓

2 Will your home have multiple bathrooms operating at the same time?

YES

Simultaneous usage

Multiple bathrooms, kitchen, laundry and outdoor taps used together can cause pressure drops.

NO ↓

3 Is your villa three floors or more?

YES

Vertical challenge

Higher floors require stronger and more stable pressure to deliver comfort.

NO ↓

4 Planning additional water features or outdoor usage?

YES

Extra demand

Pools, outdoor showers, garden irrigation and maid's room increase overall demand.

Pressure planning is recommended

Your renovation has multiple factors that can impact water pressure. Plan your water system early.

OR

Standard pressure system may be enough

Your current plan indicates lower water demand. Still, confirm with a professional before finalizing your system.

IF YOU ANSWERED YES TO TWO OR MORE QUESTIONS

A smart pressure solution like SCALA2 is designed for homes like yours — it maintains constant pressure across all outlets, on every floor, even when multiple taps are in use.



Constant
pressure



Whisper
quiet



Energy
efficient



Built for
UAE homes

AVOID THIS MISTAKE

Common renovation mistakes

Four mistakes account for most of the water complaints that surface after a villa renovation. All four are decisions, not defects — which means all four are avoidable.

HOW A BEAUTIFUL BATHROOM BECOMES A DISAPPOINTING ONE

Beautiful
rain shower

Premium
fittings

Luxury
bathroom



Poor pressure



Disappointing experience

Pressure planning should happen before finishes.

Every element above is chosen for the experience it promises. Pressure is what decides whether the promise is kept.

Mistake #1 Choosing bathroom fixtures before checking water performance

The fixture list sets the demand. Choosing it first means the system is asked to meet a target nobody measured.

Mistake #2 Installing rain showers without confirming available pressure

A rain head needs volume and steady pressure to feel like a rain head. Without both, it becomes an expensive drip.

Mistake #3 Replacing tiles while keeping an undersized pump

The room is new, the experience is not. The one component that decides how water feels was the one left untouched.

Mistake #4 Assuming a new villa automatically has an adequate pressure system

New means new. It does not mean sized for three floors, two rain showers and irrigation running together.

KNOW YOUR DEMAND BEFORE YOU DESIGN

Luxury bathroom planning checklist

Every yes below adds water demand to your villa. Tick honestly — this is the fastest way to know whether pressure is a detail or a deciding factor.

☐ **Rain shower?**
Uses two to three times the water of a standard head.

☐ **Twin vanity?**
Two outlets designed to be used at the same moment.

☐ **Garden irrigation?**
The longest pipe run on the plot, often on a timer.

☐ **Pool?**
Top-up and wash-down demand on top of the household.

☐ **Three floors?**
Vertical lift the pump has to overcome, every time.

☐ **Maid's room or guest suite in daily use?**
Another bathroom drawing at peak hours.

YOUR SCORE

If you answered yes to two or more, water pressure planning becomes critical.

They are all asking the same system for more at the same time. Decide the system now, while it is still a line on a drawing.



Plan for how you will actually live

1 Bathroom design

Plan the layout around how the space will be used. Rain heads, body jets and twin vanities all raise the water the room asks for.

2 Kitchen planning

Map the water demand of the kitchen, from the main tap to the filter, the dishwasher and any pot filler, before the joinery is fixed.

3 The most overlooked renovation decision

Water pressure planning — why it comes first

Most homeowners choose tiles before they understand how their water will perform. Pressure problems then appear after the renovation, when they are hardest and most costly to fix.

Fixture selection

What you install decides how much water the room asks for.

Pipe layout

Routing and diameter decide what actually arrives at the outlet.

Pump performance

The pump decides whether pressure holds when demand rises.

Your bathroom experience depends on all three — and changing the pump after renovation often requires additional cost and disruption. Decide how the whole house will hold pressure before the finishes are locked in.

4 Questions to ask your contractor

- ? Will pressure remain stable across all bathrooms at the same time?
- ? What happens when several taps and an appliance run at once?
- ? What booster system are you recommending, and why that one?
- ? What would it cost to change the pump after the work is finished?

DECISION REQUIRED — BEFORE SELECTING BATHROOM FIXTURES

Have you confirmed:

- ☐ Water pressure ☐ Pump capacity ☐ Number of simultaneous outlets

5 Acoustics & quiet

Quiet pumps, soft-close joinery and considered insulation keep a renovated home calm. Specify for silence, not just looks.

6 Lighting & ambience

Layer task, ambient and accent lighting per zone, and plan the circuits and dimming before the ceilings are closed.

7 THE MODERN WATER SYSTEM

Specify it like any other system

Increasingly specified in villa renovations because it supports demanding bathrooms, multiple outlets at once, quiet operation and energy-efficient performance. The Grundfos SCALA2 delivers all four in one compact unit — it belongs on the brief, not the snag list.

Constant pressure, however many taps

If your audit showed a minor or significant drop, the issue is the system behind the taps. Traditional single-speed pumps run at one speed whatever the demand, which causes the fluctuations you felt and sends vibration through the pipework.

THE SOLUTION

The Grundfos SCALA2 installs on your main line or tank outlet. Its variable-speed drive senses demand continuously. When a second or third tap opens, it scales its output instantly to hold constant pressure across every fixture at once.

- **Senses demand.** An integrated sensor reads pressure many times a second.
- **Holds the setpoint.** Speed rises and falls to keep pressure constant, 1.5 to 5.5 bar.
- **Runs quietly.** Around 44 dB(A) in normal use, with soft starts and stops.



GRUNDFOS SCALA2

1.5–5.5 bar

PRESSURE SETPOINT

8 m

SELF-PRIMING LIFT

45 m

MAXIMUM HEAD

BEFORE VS AFTER — WHAT PRESSURE PLANNING CHANGES

Planned without pressure

- × Fixtures chosen first, system last
- × Rain shower fades when a second tap opens
- × Upper floors weaker than the ground floor
- × Pump swapped after handover, at your cost
- × Walls and finishes reopened to reach the pipework

Planned with pressure

- ✓ Demand mapped before the fixture list is locked
- ✓ Every outlet holds pressure at peak hour
- ✓ Top floor performs like the ground floor
- ✓ Booster specified once, on the original brief
- ✓ Finishes stay untouched after handover

Ready to finalize your water system?

Before construction begins, make sure your contractor has specified a pressure system capable of supporting your home's future water demand — not just the demand you have today.

SPECIFY SCALA2
on your contractor brief